

MIND THE GAP: WHY LONDON'S NET-ZERO AMBITIONS DEPEND ON ITS 'HIDDEN' HOUSING DEFICIT.

London's official net-zero strategy sets an ambitious target for the capital to become a zero-carbon city by 2030. To reach this, the city must substantially cut emissions across buildings, transport, and energy supply. Current planning policies focused on new developments have successfully saved tens of thousands of tonnes of carbon; however, what is often neglected is the vast "avoidable" emissions locked within London's existing housing stock.

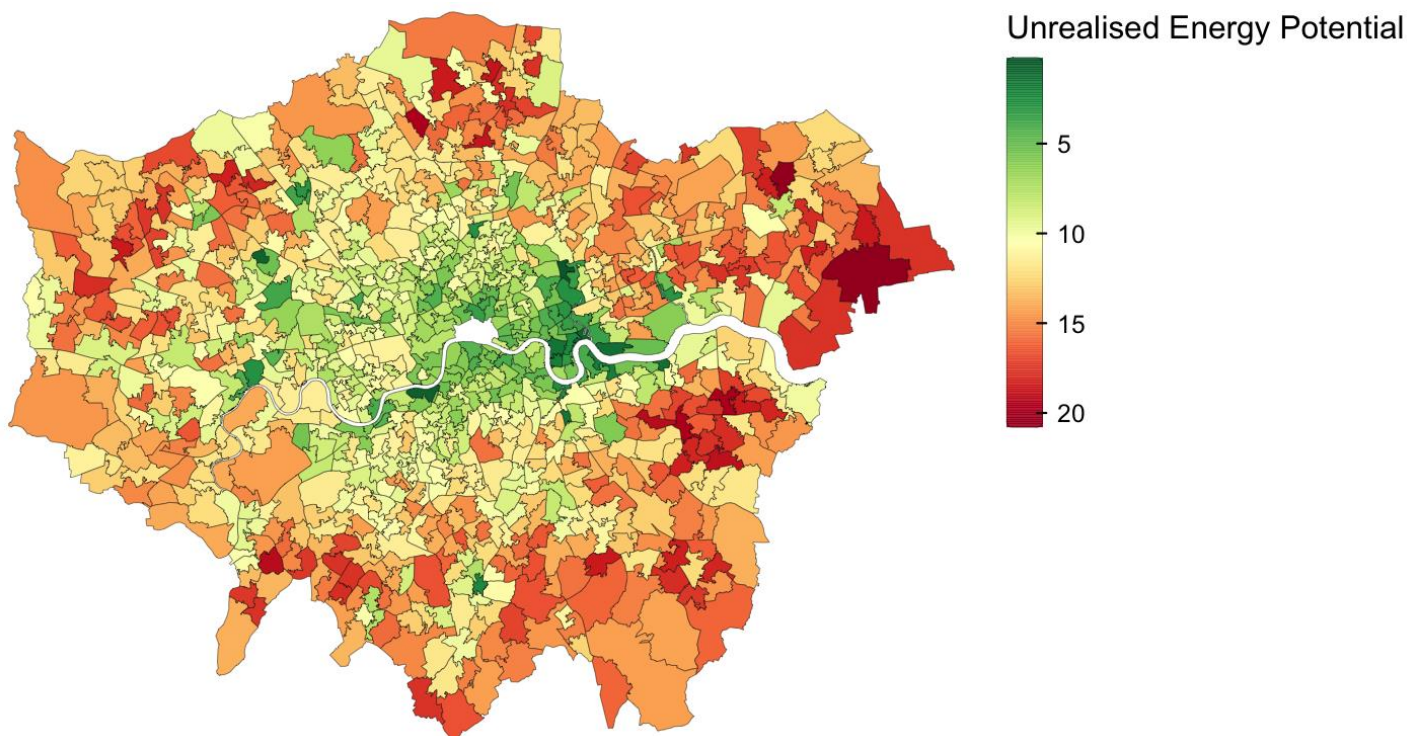
New analysis of 3.6 million Energy Performance Certificates (EPCs) reveals that the average London home is currently trapped in an Energy Efficiency Gap (EEG) of 11.4 points — the untapped potential for energy-effective retrofiting. Not only does this gap represent a carbon footprint – equivalent to a whopping 456.28 tonnes of CO₂ per annum – but also a "hidden tax" on Londoners during an ongoing cost-of-living crisis.

The spatial distribution of this "unrealised energy potential" is far from uniform. Aggregating EPC records to the neighbourhood (MSOA) level reveals a striking geographic pattern: outer-city boroughs such as Bromley and Havering have wider efficiency gaps than inner-city boroughs like Tower Hamlets.

Mapped: The hurdle to a Net Zero London

A neighbourhood-level view of 'Unrealised Energy Potential' shows that London's 2030 zero-carbon targets face a vastly different reality from one street to the next.

Average Energy Efficiency Gap by MSOA

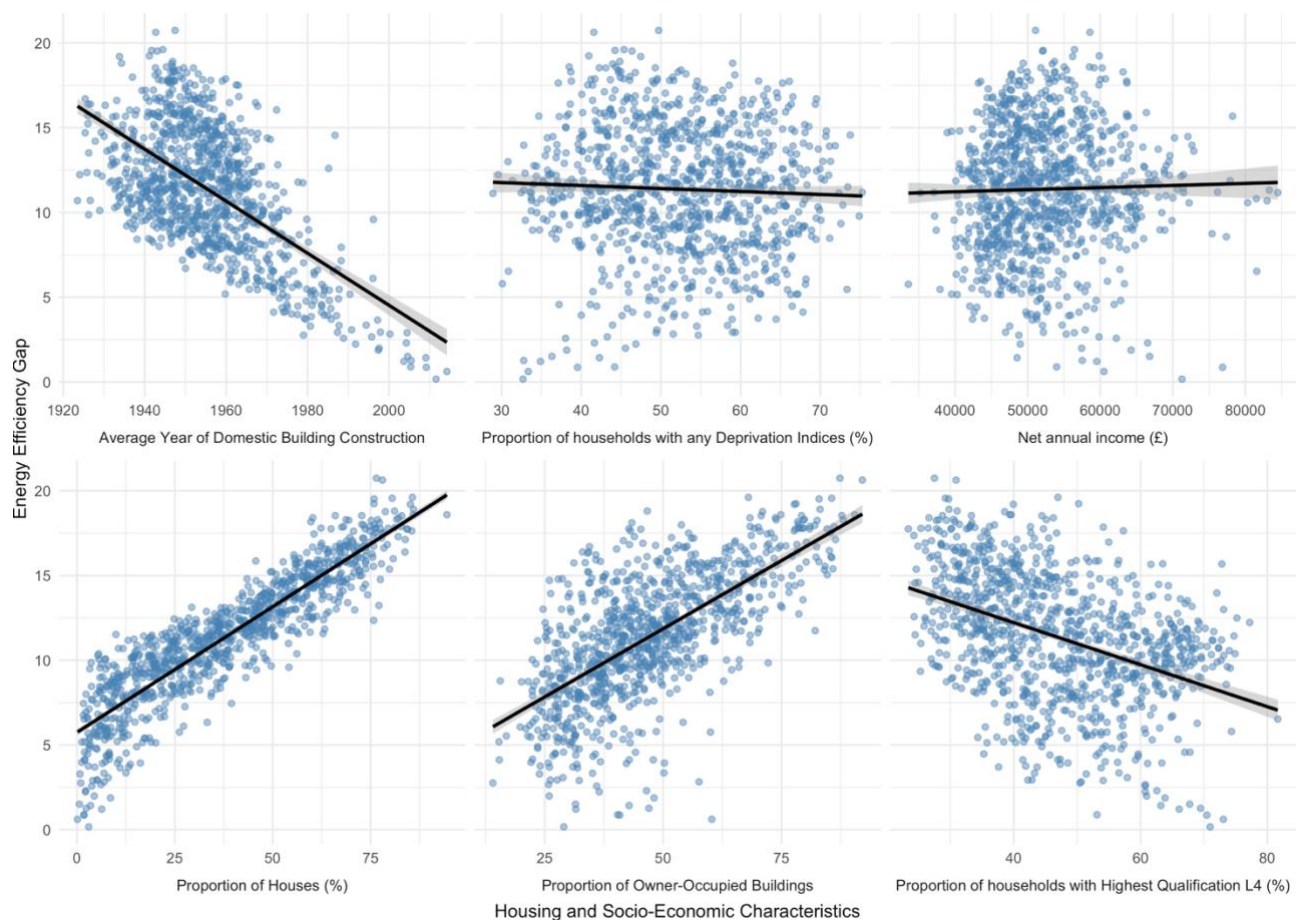


This isn't merely a result of local policy, but a reflection of London's architectural history. Property age is one of the strongest predictors of this gap. [Victorian and Edwardian terraces](#), while culturally iconic, were built long before modern insulation, glazing or heating systems were common. For these heritage-heavy neighbourhoods, reaching "potential" efficiency requires complex, expensive interventions that modern apartment blocks simply do not need.

Another clear pattern is the role of housing type. London is often described as a city of flats, but that description only applies in parts. Many outer-London neighbourhoods are still dominated by houses — and these areas tend to have much larger energy efficiency gaps. Why? Houses generally have more external walls, more exposed roof area and greater heat loss than flats. They are also harder and more expensive to retrofit comprehensively. When we account for other factors, neighbourhoods with a higher proportion of houses consistently show larger unrealised efficiency potential.

Do socio-demographics influence the energy efficiency gap? Not when housing characteristics constrain the improvements that are possible

Relationships between neighbourhood Energy Efficiency Gaps and building age, property type, tenure and household deprivation, income and qualification



One of the most revealing findings from the data is the role of housing tenure. While social housing providers often operate centralised, standardised retrofitting programmes, private homeowners face a very different set of hurdles. The data suggests a clear “tenure trap”, where neighbourhoods with high proportions of owner-occupiers tend to exhibit larger energy efficiency gaps. At first glance, this appears counter-intuitive: homeowners stand to benefit directly from lower energy bills. Yet retrofitting is expensive, disruptive and often uncertain in terms of payback, particularly in older or listed properties. As a result, many owner-occupiers appear to delay or avoid major upgrades altogether. This pattern reflects a familiar problem in urban policy — a form of [split incentive](#) operating within owner-occupied housing itself. In contrast, parts of London with large social or private rental sectors often show evidence of past waves of standardised upgrades driven by regulation or landlord investment. While overall efficiency levels in these homes remain far from ideal, their unrealised efficiency gap is often smaller. Substantial work remains, however, if London is to meet the Mayor’s ambition of becoming a net-zero capital by 2030.

Socio-economic factors do matter, but less than is often assumed. Neighbourhoods with higher proportions of residents holding degree-level qualifications tend to have smaller energy efficiency gaps, suggesting that awareness, information and engagement play a role in whether efficiency potential is realised. By contrast, broader measures of deprivation and household income show little direct relationship with the gap once housing characteristics are accounted for. This does *not* mean deprivation and income are irrelevant — but rather that the physical housing stock constrains what is possible, regardless of income. In other words, even well-resourced households struggle to close the efficiency gap when they live in hard-to-retrofit homes.

Around 40% of the variation in neighbourhood energy efficiency gaps can be traced to differences between boroughs. This suggests that local policies, planning rules, conservation constraints, retrofit programmes or institutional capacity play a major role. Some boroughs consistently “over-perform”, achieving smaller gaps than expected given their housing stock. Others systematically lag behind. This suggests that local council policies, available grants, and even local planning restrictions on heritage buildings play a massive role in whether a home can take advantage of high energy efficiency potential. These geographic differences matter because energy efficiency policy in England is increasingly devolved, with local authorities expected to play a central role in delivery. They also pose significant challenges for national policies, such as the [Social Housing Decarbonisation Fund](#), which must navigate 32 different local policy environments in London alone.

The data makes it clear: to close the gap, we must move beyond a “one-size-fits-all” approach. London needs targeted interventions that recognise the specific architectural and socio-economic realities of each borough. Neighbourhoods dominated by older houses face fundamentally different challenges from dense flat-based areas. Second, focusing solely on income or deprivation misses the point. While affordability matters, the physical structure of housing is the main constraint on progress. Retrofit support needs to be designed around building form and age, not just household characteristics.

Third, local authorities matter — a lot. The large differences between boroughs suggest that policy design, institutional capacity and local constraints can either unlock or suppress retrofit potential. Learning from high-performing boroughs may be just as important as national funding levels.

TECHNICAL APPENDIX

1. Data

1.1 Energy Performance Certificates

This analysis uses the UK domestic Energy Performance Certificate (EPC) open dataset for all London Local Authorities (excluding the City of London, which represents a single MSOA). EPC data provides standardised, publicly available information on modelled energy efficiency and environmental performance of buildings, as well as key building characteristics such as property type, tenure and construction age bands.

The main outcome variable of this study is the Energy Efficiency Gap (EEG):

$$\text{Gap}_{\text{Energy Efficiency}} = \text{Potential}_{\text{Energy Efficiency}} - \text{Current}_{\text{Energy Efficiency}}$$

By utilizing this specific metric instead of current energy efficiency, the analysis avoids circularity with the Standard Assessment Procedure (SAP) model inputs.

Cleaning steps:

- Removing duplicates (keeping the most recent inspection per building, yielding ~3.6 million records)
- Converting construction age bands to numeric midpoints
- Aggregating EPC data at MSOA level using Postcode to MSOA lookup data.

1.2 Socio-Economic Data

EPC data were integrated with MSOA-level socioeconomic variables (local qualifications, deprivation indices, household income, and median property values). Tenure and property type fields were aggregated into proportions representing building stock within each MSOA.

2. Exploratory Data Analysis

2.1 Energy Efficiency Gap

The distribution of MSOA-aggregated EEG is approximately normal (mean 11.4, median 11.2, Figure 1), justifying linear regression modelling.

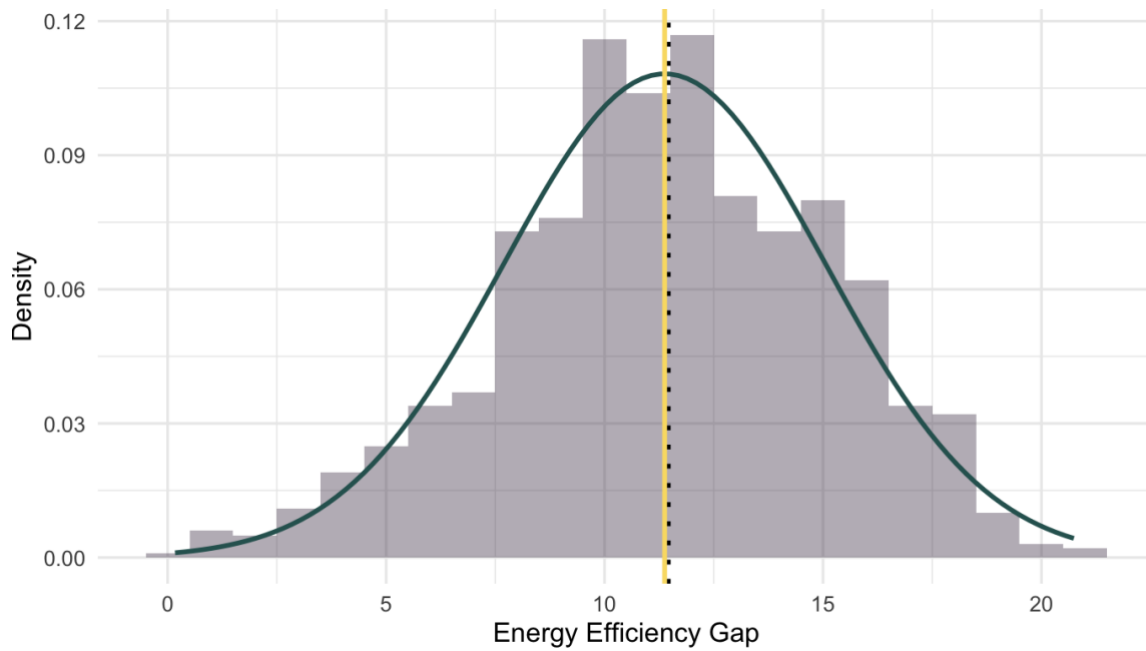


Figure 1. Distribution of London Energy Efficiency Gap data aggregated at MSOA-level

2.2 Predictors

Ternary plots were created for property types (Figure 2) and tenure (Figure 3) to understand the composition of London's housing stock visually. The vast majority of domestic buildings in London are either houses or flats, with other property types (maisonettes, bungalows, and park homes) contributing minimally to the composition of housing stock (Figure 2).

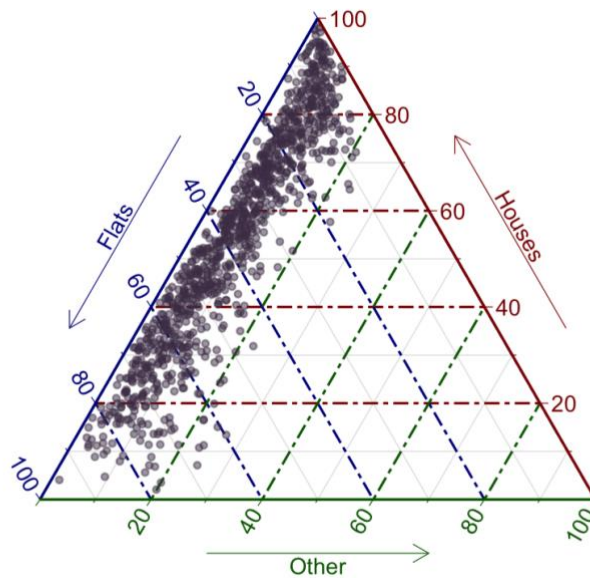


Figure 2. Ternary plot of housing tenure in London MSOAs

Tenure is more varied, with few MSOAs being dominated entirely by owner-occupied, socially rented or privately rented housing (Figure 3). For simplicity, subsequent analyses

includes only the proportion of owner-occupied housing and the proportion of houses as indicators of London's MSOA domestic property stock.

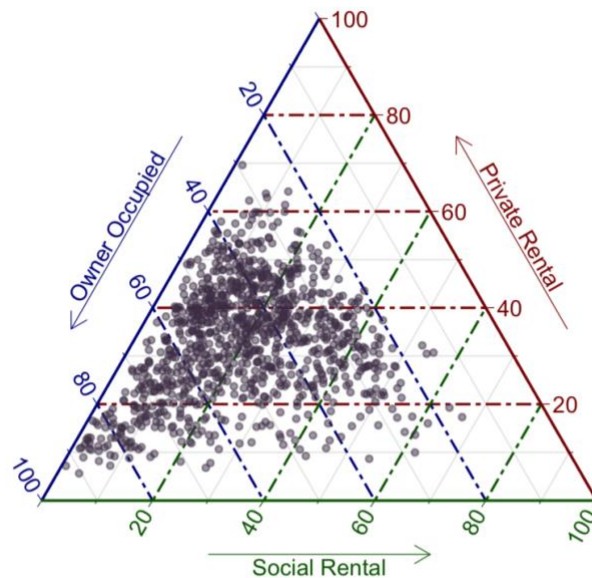


Figure 3. Ternary plot of domestic property types in London MSOAs

Energy Efficiency Gap (EEG) is strongly associated with a combination of built-environment and socio-economic factors (Figure 4).

- **Primary Predictors:** Scatterplots reveal strong positive correlations between the EEG and the proportion of houses, owner-occupied properties, and average building age (older buildings exhibit larger gaps).

2.3 CO₂ Emissions Calculation

For each building, “wasted” CO₂ emissions were calculated (potential minus current emissions in kg/m²/annum × floor area). Total untapped emissions = 456,276,789 kgCO₂ per annum.

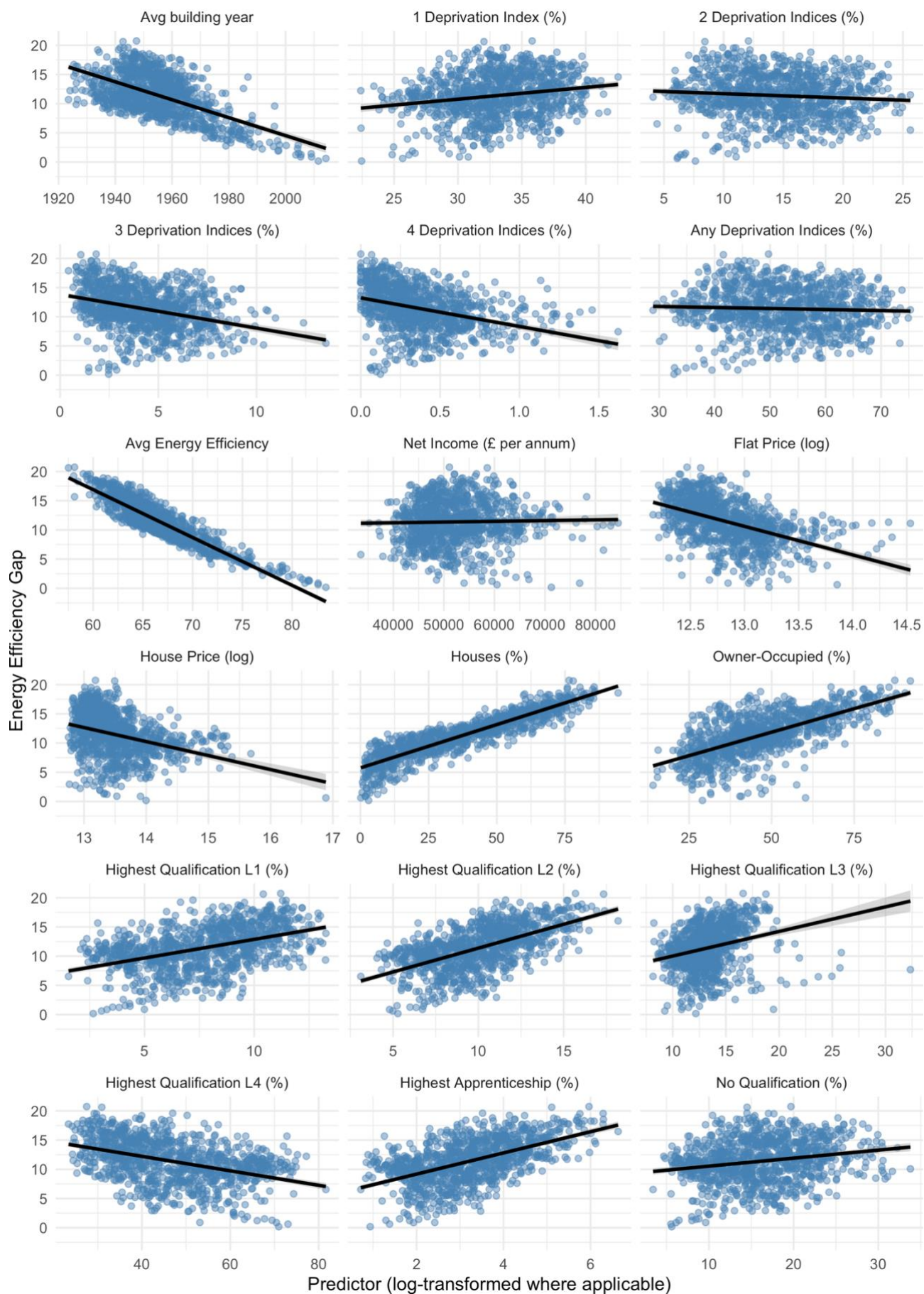


Figure 4. Faceted Scatterplots of Energy Efficiency Gap vs Predictors

- **Socio-Economic Factors:** A notable negative relationship exists between the EEG and Level 4 qualifications, suggesting that higher education levels in an MSOAs correlate with a reduced energy efficiency gap. Although other qualification-related variables also have strong relationships with EEG, Level 4 shows a much wider range of proportions across London MSOAs (roughly between 30% and 80%), meaning its effect on EEG is more pronounced. Although the proportion of households with at least one deprivation index does not seem to have an effect on EEG, we include this variable in the analysis given its social relevance.

3. Regression Modelling

3.1 Ordinary Least Squares (OLS) models

We use OLS regression to model EEG and our chosen predictor variables hierarchically (Table 1). Property type, tenure and construction age have a significant influence on EEG, with property type being the main determinant of the gap (causing R^2 to jump from 0.44 in model 1 to 0.82 in model 2). While the model fit does not change after model 3, qualifications still have a significant influence on EEG, unlike deprivation.

Table 1. OLS Models Outputs for EEG and Predictor Variables

	Model 1	Model 2	Model 3	Model 4	Model 5
(Intercept)	3.81 ***	4.91 ***	180.04 ***	199.47 ***	197.50 ***
prop_owner	0.16 ***	0.03 ***	0.03 ***	0.05 ***	0.05 ***
prop_house	—	0.14 ***	0.12 ***	0.10 ***	0.10 ***
year_avg	—	—	-0.09 ***	-0.10 ***	-0.10 ***
q_4_prop	—	—	—	-0.04 ***	-0.03 ***
dep_prop	—	—	—	—	0.01
R^2	0.44	0.82	0.93	0.93	0.93

It is important to note that OLS regression assumes that MSOA-level observations are independent. In practice, this assumption is unlikely to hold, as MSOAs are hierarchically nested within Local Authorities (LAs), which are shared administrative and policy environments. This nesting may result in clustering of both unrealised energy efficiency gaps and their relationships with predictor variables, and a random effects model may be required to account for clustering effects.

3.2 Random Effects Models

The variance components model ($EEG \sim 1 + (1 | \text{Local Authority})$) showed that the EEG varies by 5.782 unites on average between LAs and 8.141 units within LAs. This corresponds to an Intra-class Correlation Coefficient (ICC) of 0.42, meaning 42% of the total variation in energy efficiency gaps is explained by differences between LAs. This,

combined with high variation in actual EEG intercepts between LAs (Figure 5), justifies exploration of more complex linear mixed effects models.

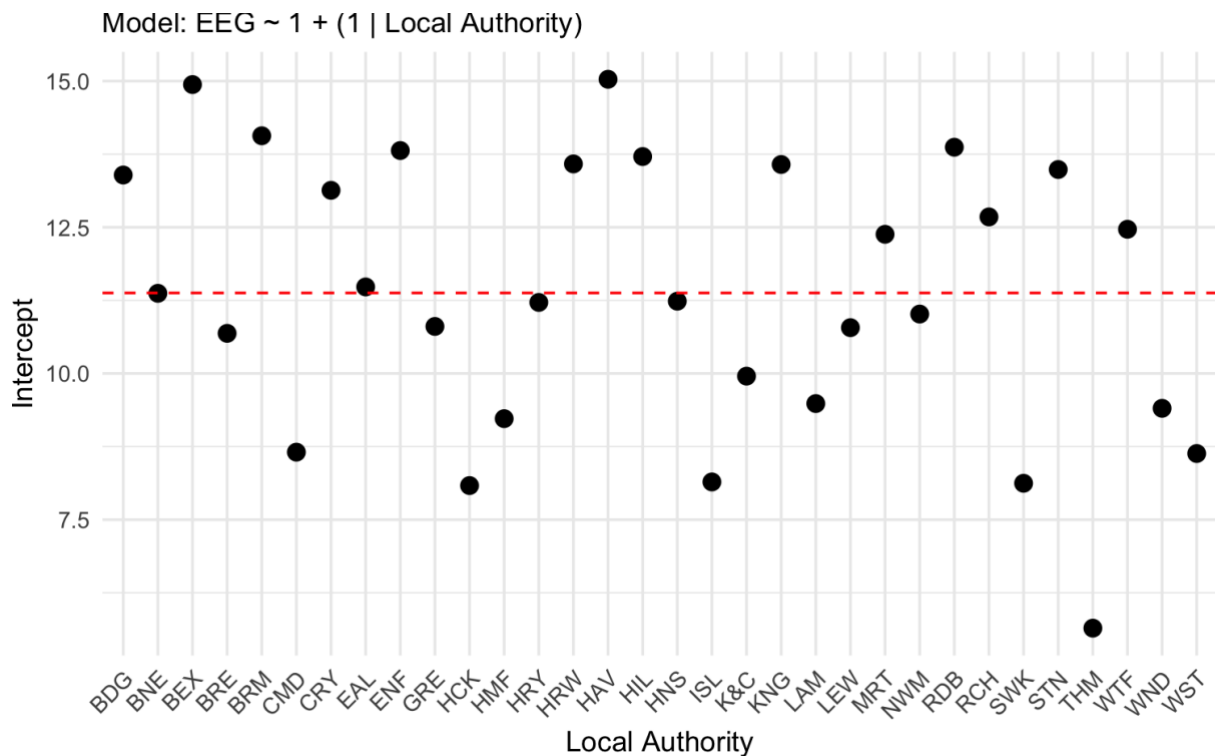


Figure 5. Actual Energy Efficiency Gap Intercepts by Local Authority

The addition of housing type as a fixed effect predictor decreases the residual variance from 8.141 in the variance components (null) model to 2.096, and decreased the REML criterion from 5038 to 3664. This indicated a drastic improvement in model accuracy.

Random slope effects for property type and qualification metrics appear to differ drastically between LAs (Figure 6). However, model summaries show that even though REML criterion dropped statistically for some of the variables (property type and tenure), the slope variance for all four variables is negligible. Additionally, the correlations between the slopes and intercepts are all close to +1. Thus, the visual differences in variable vs EEG slopes between LAs in figure x are not statistically reliable, and adding a random slope term to the model is not worth the additional complexity.

Thus, our final model is:

$\text{gap_avg} \sim \text{prop_house} + \text{prop_owner} + \text{q_4_prop} + \text{age_avg} + (1|\text{london_inner_outer}/\text{LA})$

Only 15.7% of the unexplained variation is attributable to differences between Local Authorities rather than MSOA-MSOA differences (Table 2). Property type and property age are most influential predictors of the EEG, with positive and negative correlations, respectively (Table 2). A key limitation is that the main predictors — property type,

construction year, and tenure — cannot be changed. A building cannot, for example, be converted from a house to a flat to reduce its energy efficiency gap. The model is therefore best used to understand patterns and trends in EEGs and guide neighbourhood-level policy, rather than as a prescriptive tool.

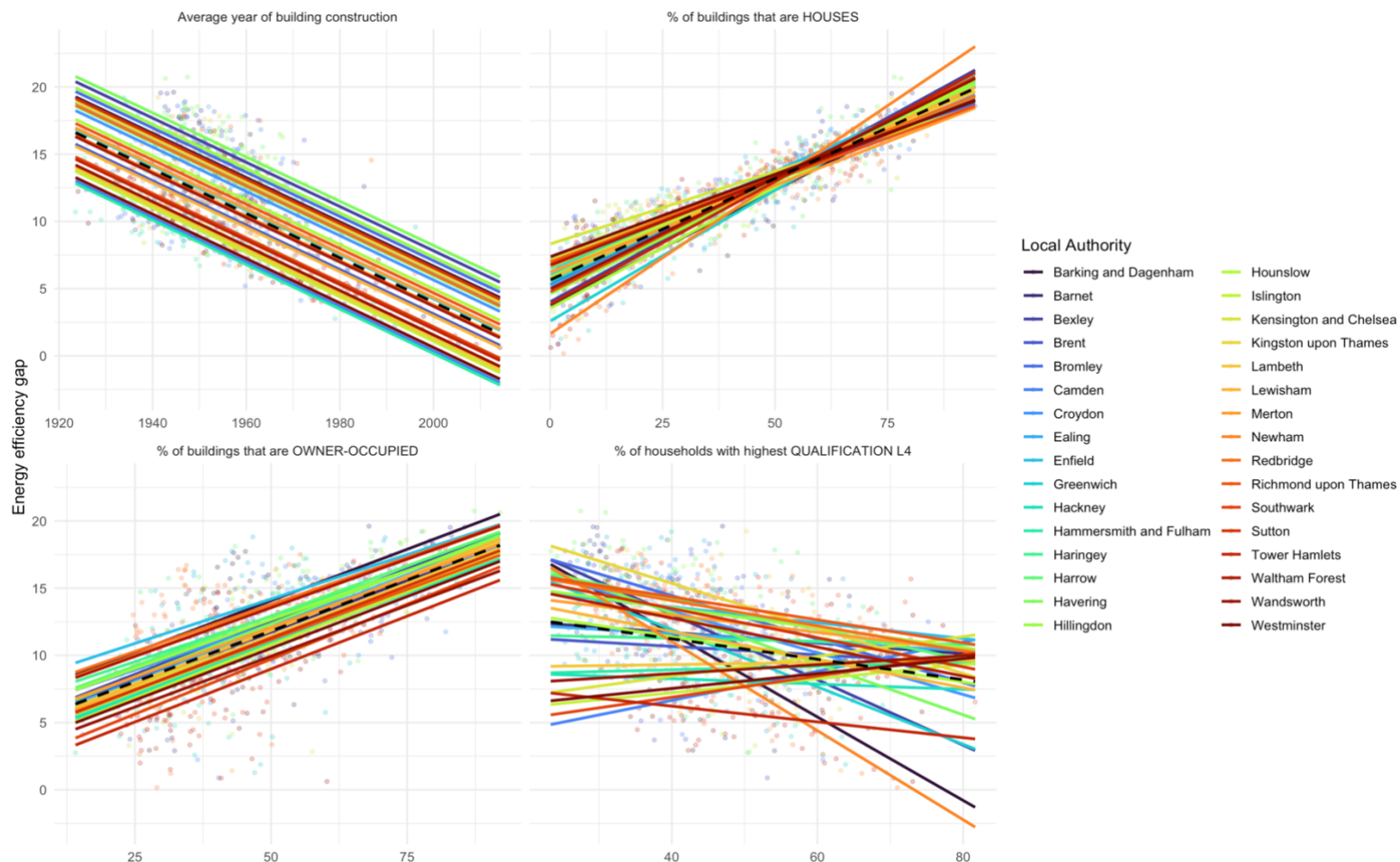


Figure 6. Random Slope Effects of building age, property type, tenure and qualification on the energy efficiency gap by Local Authority

Model assumption checks indicate an overall good fit and no major violations (Figure 7). The caterpillar plot shows the estimated Local Authority-level random intercepts from our final linear mixed-effects model (Figure 8). Boroughs whose error bars do not cross the vertical zero line are statistically significant “over” or “under” performance. This indicates LAs where some other factor not included in our model is important.

Table 2. Final Linear Mixed Model Effects Outputs

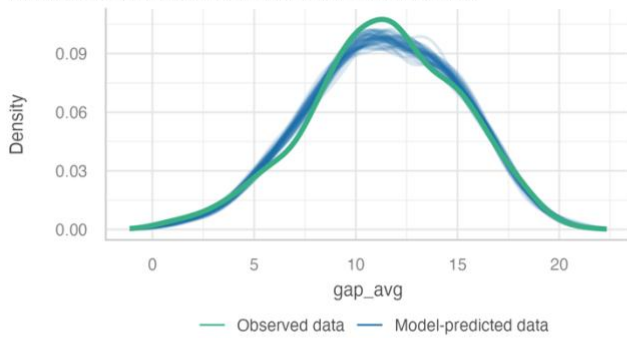
Fixed effects	Variable/Group	Estimate	Std Error	t-value
	Intercept	199.45	4.90	40.71
	Proportion of houses	0.098	0.003	34.74
	Proportion owner-occupied	0.045	0.004	12.61
	Proportion Q4 deprivation	-0.030	0.004	-7.95
	Average building age	-0.098	0.002	-39.89
Random effects		Variance	SD	
	Intercept: LA within region (inner/outer)	0.146	0.382	—
	Intercept: Inner vs outer London	0.000	0.000	—
	Residual MSOA-to-MSOA	0.783	0.885	—

3. Limitations

- **EPC Latency:** Homes without recent transaction (sale or rental) may have unrecorded retrofits, potentially overestimating the EEG.
- **Modifiable Areal Unit Problem:** Aggregating to MSOA level may mask significant street-level variations.
- **Spatial Autocorrelation:** While the LME accounts for LA-level clustering, residual spatial dependency may remain, suggesting potential to do a Geographically Weighted Regression (GWR).

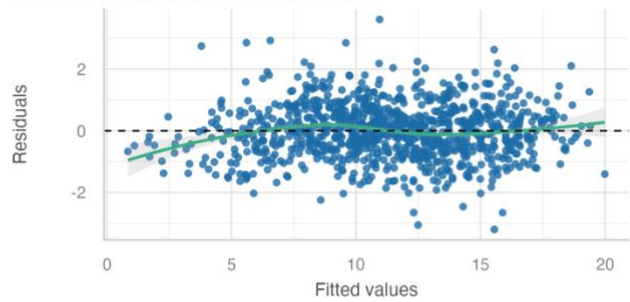
Posterior Predictive Check

Model-predicted lines should resemble observed data line



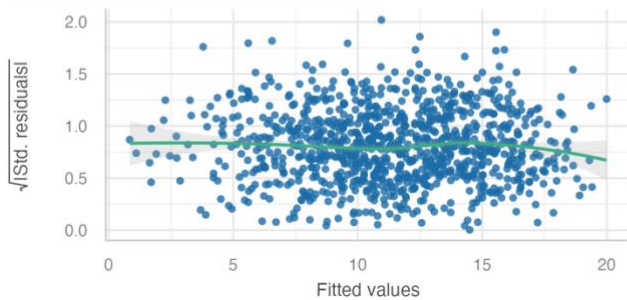
Linearity

Reference line should be flat and horizontal



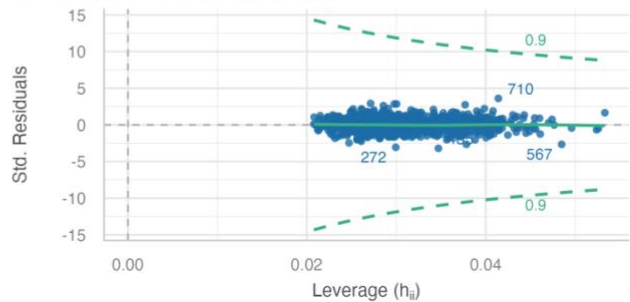
Homogeneity of Variance

Reference line should be flat and horizontal



Influential Observations

Points should be inside the contour lines



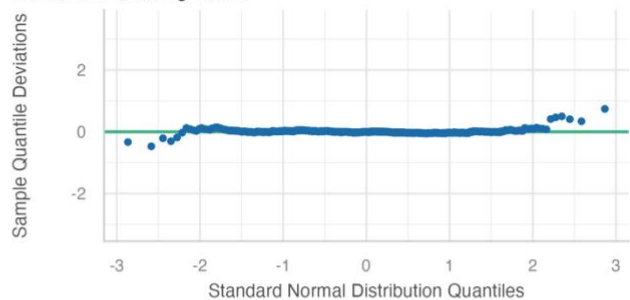
Collinearity

High collinearity (VIF) may inflate parameter uncertainty



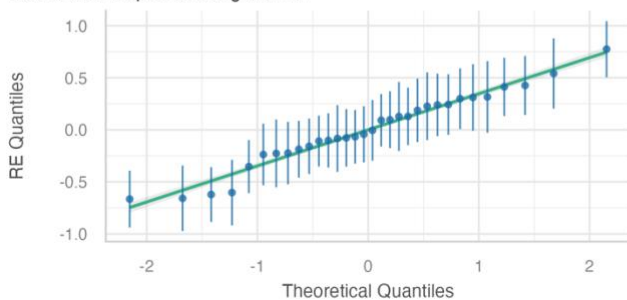
Normality of Residuals

Dots should fall along the line



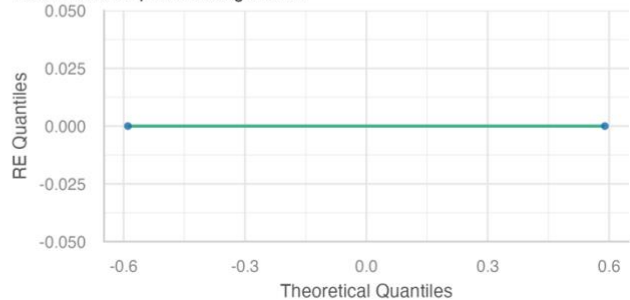
Normality of Random Effects (LAD23NM:london_inner_outer)

Dots should be plotted along the line



Normality of Random Effects (london_inner_outer)

Dots should be plotted along the line



Graph 7. Graphs to check the assumptions of our final linear mixed effects model

